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MS24129

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI
END-SEMESTER 2025-2026
IDC214: Quantitative Estimates in Soft Matter

FULL MARKS 50 DURATION 3 HR

Instructions: Please finish one entire section together at one place. DO NOT intermix Sections.

Section A: (20 marks)

Q1. Derive the equation of motion for a particle of mass ***m*** attached to a spring of spring constant ***k*** and moving in a viscous fluid experiencing a linear drag force $F_d = -b.v$ **[Hint.** Apply Newton's 2nd law of motion] **3**

B. Assuming the particle is micron-sized and suspended in water, simplify the equation of motion using the low Reynolds number approximation and Stokes drag. Comment on whether the motion is underdamped, overdamped, or critically damped. **3**

C. Using the simplified equation obtained in part B, derive the relaxation time of the micron-sized particle and discuss its dependence on ***k*** and ***b***. **2**

Q2. In a sentence or two, explain the physical meaning of the time autocorrelation function. **2**

Q3. What is the Stokes drag force acting on a spherical particle? **2**

Q4. Write the Einstein relation connecting the diffusion coefficient D and friction coefficient γ. **1**

Q5. State two key differences between STORM and PALM. Name one additional super-resolution imaging technique other than STED, STORM, and PALM. **2+1**

Q6. Define sedimentation equilibrium. State the condition required to attain sedimentation equilibrium, and derive the resulting concentration profile. **2+2**

Section B: (20 marks) (Answer all questions of Section B in **Tabular Format** using a single consolidated table, with one row per question. Write the **most suitable** tool. **Naming** any related technique will **NOT** be accepted.)

Q7. Your experiment requires ~1 nm localization accuracy for tracking a molecule moving on a surface at a speed of the order of nm/s. What would be your measurement technique for capturing such a motion and why (not more than two sentences)? What parameter primarily determines the localization accuracy for this? **1 + 2 + 1**

Q8. A biomolecule exerts forces in the sub-piconewton range on its interacting partner. Which experimental tool would you choose for accurate force estimation and why? What is the typical range of forces that can be measured using your selected tool? **1 + 2 + 1**

Q9. Two subunits of a protein undergo periodic relative motion. How would you quantitatively measure periodic nanometer-scale changes in the inter-subunit distance

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and determine the oscillation period? Explain why? What is the spatial range and temporal resolution of detection of the selected technique? **1 + 2 + 1**

Q10. You want to image the actin cytoskeleton in living cells with super-resolution, without labelling with organic or inorganic fluorophores. Which imaging technique would you choose and why? What is the spatial resolution of the selected technique? **1 + 2 + 1**

Q11. Which experimental technique would you use to monitor DNA supercoiling force and relaxation time without applying optical torque? What is the typical force resolution of the selected technique? why? **1 + 2 + 1**

Section C: (10 marks) (All answers in this section should be graphical. Use Figure legends and figure captions to describe the figure. Figure caption should not exceed more than 3 sentences.)

Q12. Using a suitable mechanical analog (e.g., Maxwell), illustrate the behavior of a simple viscoelastic material. **2**

Q13. Draw *schematic shear* stress versus shear rate curves for Newtonian, shear-thinning, and shear-thickening fluids. **3**

Q14. Sketch the mean square displacement (MSD) as a function of time for a Brownian particle diffusing in two dimensions. **2**

Q15. Illustrate schematically the difference between laminar and turbulent flow in a pipe, and state the approximate Reynolds number ranges corresponding to each regime. **2 + 1**